Description

Illustration





Figure caption

1. Close up texture of slate. 2. Traditional roofing slates being laid on a roof. Images used under license from Shutterstock.com

The material

Slates are formed by the deposition of clay and mud, consolidated by pressure. Their most striking features are their ability to be cleaved, producing flat slabs or sheets, and their impermeability to water in a direction normal to the cleavage plain. Because of this, slate has been used for centuries for roof tiles, pavings and floors when laid directly on bare earth. Slate is exceptionally durable and weather resistant: if maintained, slate roofs last for hundreds of years. The fine texture and uniform subdued coloring leads also to the decorative use of slate, in stair treads, signs and grave stones.

Compositional summary

Slate is a form of shale, a complex alumino-silicate.

General properties

Density	2.6e3	-	2.9e3	kg/m^3
Price	* 0.513	-	0.672	GBP/kg
Date first used	-10000			

Material form that data applies to

Bulk	✓

Building system

Superstructure	✓
Enclosure	✓
Interiors	✓
Services	✓

Mechanical properties

Young's modulus		60	-	90	GPa
Shear modulus	*	20	-	30	GPa
Bulk modulus	*	50	-	80	GPa
Bending modulus	*	60	-	90	GPa
Poisson's ratio	*	0.22	-	0.3	
Yield strength (elastic limit)	*	15	-	30	MPa
Tensile strength	*	15	-	30	MPa
Compressive strength		120	-	175	MPa
Bending strength	*	20	-	50	MPa
Elongation	*	0.02	-	0.05	% strain
Hardness - Vickers		22	-	60	HV
Fatigue strength at 10^7 cycles	*	10	-	12	MPa
Fracture toughness	*	0.4	-	1.1	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.001	-	0.003	

Thermal and combustion properties

Thermal conductor or insulator?		Poor insulator			
Thermal resistivity		0.278	-	0.625	m.°C/W
Thermal expansion coefficient		10	-	12	µstrain/°C
Specific heat capacity		850	-	890	J/kg.°C
Melting point	*	1.2e3	-	1.6e3	°C
Maximum service temperature	*	350	-	500	°C
Minimum service temperature		-50	-	-30	°C
Flammability		Non-flammable			
Emissivity		0.9			

Hygro-thermal properties

Water absorption		0.01	-	0.1	%
Frost resistance		Very good			

Electrical properties

Electrical conductor or insulator?		Good insulator			
Electrical resistivity	*	1e12	-	1e14	µohm.cm
Dielectric constant (relative permittivity)	*	8	-	15	
Dissipation factor (dielectric loss tangent)	*	0.001	-	0.005	
Dielectric strength (dielectric breakdown)	*	5	-	12	MV/m

Optical properties

Natural color

Gray

Transparency	Opaque			
Transmissivity	0			%

Critical Materials Risk

High critical material risk?	Yes			

Acoustic properties

Sound absorption	Poor			
Sound isolation	Good			

Processability (scale 1 = impractical to 5 = excellent)

Machinability	3	-	4	

Durability

Water (fresh)	Excellent
Water (salt)	Acceptable
Weak acids	Excellent
Strong acids	Excellent
Weak alkalis	Excellent
Strong alkalis	Acceptable
Organic solvents	Excellent
UV radiation (sunlight)	Excellent
Wear resistance	Excellent
Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent

Primary material production: energy and climate change

Climate change (CO_2-eq) (virgin grade production)	* 0.0333	-	0.0368	kg/kg	
Embodied energy (virgin grade production)	* 0.524	-	0.577	MJ/kg	
Water Usage	* 3.23	-	3.57	l/kg	

Material processing: energy

Machining, grinding, embodied energy	* 28.1	-	35.5	MJ/kg	

Material processing: climate change

Machining, grinding, climate change (CO_2-eq)	* 1.84	-	2.33	kg/kg	

Recycling and end of life

Recycle	✗			
Functional recycle	✗			
Recycle fraction in current supply	6.72	-	7.42	%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
A renewable resource?	✗			

Supporting information

Design guidelines

Slate is an exceptionally stable and inert material. Slate roofs last between 30 and 100 years, depending on the climate. It can be damaged by severe frost.

Technical notes

Slate cleaves easily to give surfaces that can be extremely flat -- its use as the surface of a billiard table is an example. Its fine texture allows delicate incising and carving.

Typical uses

Roof tiles, paving, floors, stair treads, table tops (including billiard and snooker tables), chalk boards, electric panels, gravestones and other monumental signs.

Related standards

04400 Stone
04410 Rough Stone
04420 Cut Stone
04440 Flagstone
04450 Stone Veneer
04475 Slate

Further reading

Doran, D.K. (ed.) Construction materials reference book, Butterworth Heinemann, Oxford (1992)

Links

ProcessUniverse
Producers
Reference

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